IAT New Test 21

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

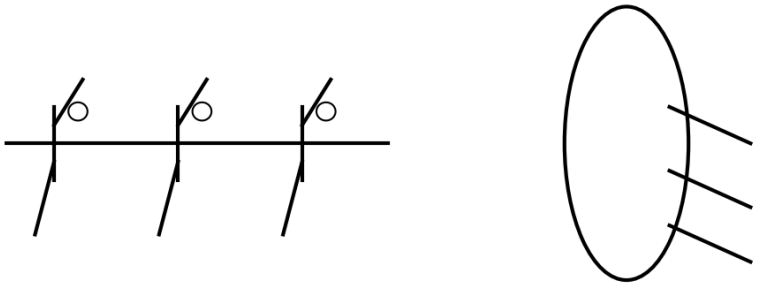
Biology

Q1 Four unicellular or simple eukaryotic preparations have these features: P has a flexible pellicle and can photosynthesise in light but use organic food in darkness; Q has a wall made of two overlapping silica-rich halves; R has cellulose plates and two differently oriented flagella; S forms a multinucleate feeding mass and later spore-bearing fruiting structures. Which ordered assignment is consistent?

A) P euglenoid; Q diatom; R dinoflagellate; S slime mould
B) P diatom; Q euglenoid; R slime mould; S dinoflagellate
C) P dinoflagellate; Q slime mould; R diatom; S euglenoid
D) P slime mould; Q dinoflagellate; R euglenoid; S diatom

Q2 Both underground structures P and Q in the figure store food. P has repeated nodes with scale leaves and axillary buds; Q bears lateral roots but has no nodes or leaf-associated buds. Which inference follows from the structural evidence rather than their shared storage function?

Underground food-storing structures P and Q



P: nodes, scale leaves and axillary buds

Q: lateral roots; no nodes

A) P and Q are both roots because they grow below ground
B) P is a modified stem and Q a modified root
C) P is a modified root and Q a modified stem
D) P and Q must be leaves because they contain stored carbohydrate

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Q3 A stipulated eukaryotic chromatin model contains 6000 bp of DNA in identical 200 bp repeats. Each repeat has one core of eight histone molecules, with 146 bp wrapped around that core and the rest as linker DNA. Ignore end effects and linker histone H1. How many core histone molecules and how many linker base pairs are present?

A) 30 molecules; 1620 bp
B) 240 molecules; 4380 bp
C) 240 molecules; 1620 bp
D) 480 molecules; 3240 bp

Q4 Mature plant parenchyma cells rarely divide in their original tissue. In a supplied culture medium they resume division and form callus. Transfer of that callus to a different growth-regulator mixture produces organised shoots. No DNA-sequence change is detected. Which interpretation is best supported?
A) The first transition includes dedifferentiation; the second includes redifferentiation under changed regulatory conditions
B) The callus must have lost all capacity to differentiate because it divided
C) Shoot formation proves a new mutation created every developmental instruction
D) The original cells completed meiosis and became gametes before forming shoots

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Q5 In an idealised central-nervous-system comparison, intervention X disrupts balance and coordination of voluntary movements while basic breathing rhythm remains. Intervention Y disrupts the normal breathing rhythm while skeletal muscles can still contract when stimulated. Which affected regions are most consistent with X and Y, respectively?
A) Spinal sensory roots and cerebral motor cortex
B) Medulla oblongata and cerebellum
C) Hypothalamus and spinal motor roots
D) Cerebellum and medulla oblongata

Q6 Developing pollen loses viability after a treatment damages the anther's nutritive inner wall layer. Moving that pollen to several otherwise compatible stigmas does not restore germination. Separate chemical treatment leaves the pollen exine resistant while removing other wall material. Which pair of assignments fits the two observations?
A) The damaged layer is the stigma; the resistant wall material is cellulose
B) The damaged layer is the tapetum; the resistant material is sporopollenin
C) The damaged layer is the ovule integument; the resistant material is pectin
D) The damaged layer is the endosperm; the resistant material is lignin

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Q7 A mother's X chromosomes bear centromere-linked markers XA and Xa; the father's chromosomes are XB and Y. Assume no crossing over between each marker and its centromere and no post-zygotic chromosome change. In separate conceptions, two XXY children have complements XA-Xa-Y and XA-XA-Y. Which maternal nondisjunction stages most directly account for the first and second complements?
A) Meiosis II and meiosis I
B) Meiosis I and meiosis I
C) Meiosis I and meiosis II
D) Meiosis II and meiosis II

Q8 Product P from Monascus purpureus competitively inhibits an enzyme in cholesterol synthesis. Product Q from Trichoderma polysporum suppresses immune responses. Which industrial-product assignments match these distinct targets?
A) P penicillin; Q streptokinase
B) P cyclosporin A; Q statin
C) P streptokinase; Q penicillin
D) P statin; Q cyclosporin A

Q9 In a stipulated ADA-deficiency gene-therapy model, functional ADA cDNA is introduced into a patient's lymphocytes, which are returned to the patient. The corrected cells make ADA but have limited persistence. Which change would most directly address the need for repeated cell infusions, if feasible?
A) Introduce a persistent corrected progenitor population that can keep supplying functional descendants
B) Remove the functional cDNA immediately after the lymphocytes enter the blood
C) Use uncorrected lymphocytes because their normal life span is longer than the cDNA sequence
D) Keep the same short-lived corrected cells but assume their temporary enzyme production permanently changes every other cell's genome

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Q10 A wasp pollinates a fig and its larvae consume some developing seeds. Measurements show that the tree produces more viable seeds with the wasp than without it, and the wasp completes reproduction only in this interaction. Which interpretation is consistent with the evidence at both scales?
A) The interaction must be parasitism at every scale because at least one seed is eaten
B) The measured whole-organism interaction can be mutualistic despite a cost to individual consumed seeds
C) The tree must receive no benefit because the wasp reproduces
D) Mutualism requires that neither partner use any material or energy supplied by the other

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Q11 Inherited disorder P reduces production of an otherwise ordinary globin-chain type. In disorder Q, a beta-globin amino-acid substitution promotes abnormal haemoglobin behaviour and red-cell sickling under low oxygen. Which distinction is most consistent with these observations?
A) P is a qualitative haemoglobin change; Q is only a reduction in chain amount
B) P and Q must both be caused solely by dietary iron deficiency
C) P fits a quantitative thalassaemia defect; Q fits the qualitative sickle-cell defect
D) P and Q cannot be inherited because mature red cells lack nuclei

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Q12 Three dated specimens have measured cranial volumes: an older Homo erectus specimen, 900 cm3; a later Neanderthal specimen, 1400 cm3; and a still later Homo sapiens specimen, 1350 cm3. Separately, human and whale forelimbs share corresponding skeletal elements despite different uses, whereas bird and insect wings share flight function without the same underlying construction. Which pair of conclusions is supported?
A) Brain volume must increase strictly in chronological order; all flight structures establish recent common ancestry
B) The sample volumes prove a direct ancestor-descendant chain; different forelimb functions exclude homology
C) The largest-brained specimen must be the ancestor of both others; bird and insect wings are homologous because both fly
D) These sample volumes do not form a strictly increasing sequence; corresponding human-whale forelimb elements support homology despite functional divergence

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Q13 A joint allows movement in several planes, including rotation, as at the shoulder. In a separate joint preparation, uric-acid crystals accompany inflammation while directly stimulated muscle fibres retain force. Which joint-type and disorder-mechanism pairing is most consistent?
A) Ball-and-socket articulation; gout-related crystal inflammation
B) Hinge articulation; loss of bone mass defining osteoporosis
C) Gliding articulation; autoimmune failure of the neuromuscular junction
D) Fixed fibrous articulation; absence of actin in every muscle fibre

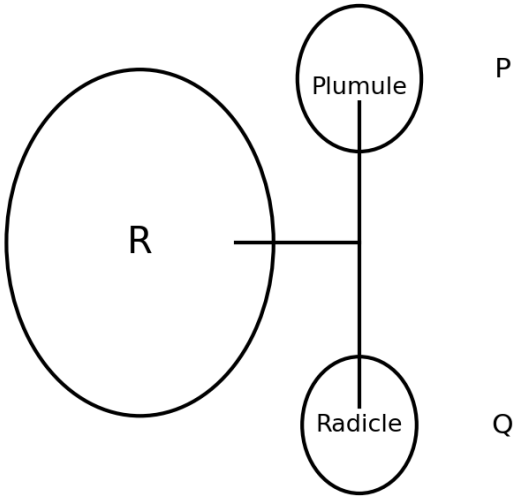
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Q14 A detergent disperses a phospholipid preparation into smaller aggregates without producing new covalent molecular species. In separate tubes, amylase produces smaller carbohydrate products from starch and a protease produces smaller peptide products from a protein. Which change is specifically physical dispersion rather than the stated covalent hydrolysis?
A) The protease reaction, because all peptide bonds are noncovalent
B) The amylase reaction, because starch is not a biomacromolecule
C) Both enzyme reactions, because catalysts cannot alter chemical bonds
D) The detergent-induced redistribution of intact phospholipid molecules

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Q15 The cereal embryo in the figure has a single large lateral structure R and two protective sheaths P and Q at opposite ends of its embryonal axis. Which assignment correctly identifies P around the plumule, Q around the radicle and R?

Cereal embryo: axis and lateral structure

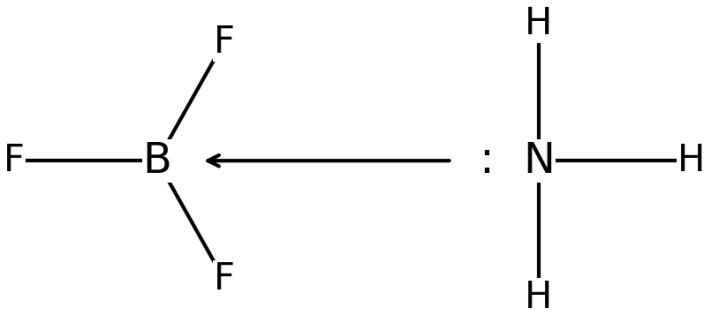


A) P coleorhiza; Q coleoptile; R endosperm
B) P coleoptile; Q coleorhiza; R scutellum
C) P scutellum; Q endosperm; R coleoptile
D) P endosperm; Q scutellum; R coleorhiza

Chemistry

Q16 In the diagram, ammonia donates its nitrogen lone pair to the electron-deficient boron of BF3, forming a single donor–acceptor adduct. Which change at boron and classification of the reaction are consistent with this model?

Lewis-pair donation: reactants



One nitrogen lone pair is donated to boron

A) Trigonal planar to linear; oxidation of boron
B) Trigonal planar to tetrahedral; no change in oxidation states
C) Trigonal planar to tetrahedral; reduction of boron
D) Tetrahedral to trigonal planar; proton transfer

Q17 A water sample contains only Ca2+ and Mg2+ as EDTA-reactive metal ions. A total-metal assay consumes 1.00 mmol EDTA, with one EDTA ligand binding one metal ion. A selective calcium assay of an equal sample consumes 0.600 mmol EDTA. What can be deduced about the first assay alone, and about magnesium after both assays?
A) The first fixes equal metal amounts; Mg is 0.500 mmol
B) The first fixes the Ca:Mg ratio at 6:1; Mg is 0.100 mmol
C) The first fixes six metal ions per ligand; Mg is 5.40 mmol
D) The first fixes only their sum; Mg is 0.400 mmol

Q18 Potassium manganate K2MnO4 and potassium permanganate KMnO4 are both coloured. Manganese has atomic number 25. In the ideal spin-only ground-state model, which comparison of their manganese moments and inference about colour is correct? Use moment = √[u(u+2)] μB for u unpaired electrons.
A) Both √3 μB; every coloured ion must have an unpaired electron
B) 0 and √3 μB; increasing oxidation state must increase d-electron count
C) √3 μB and 0; colour alone does not establish an unpaired ground-state electron
D) Both 0; every oxyanion must be diamagnetic

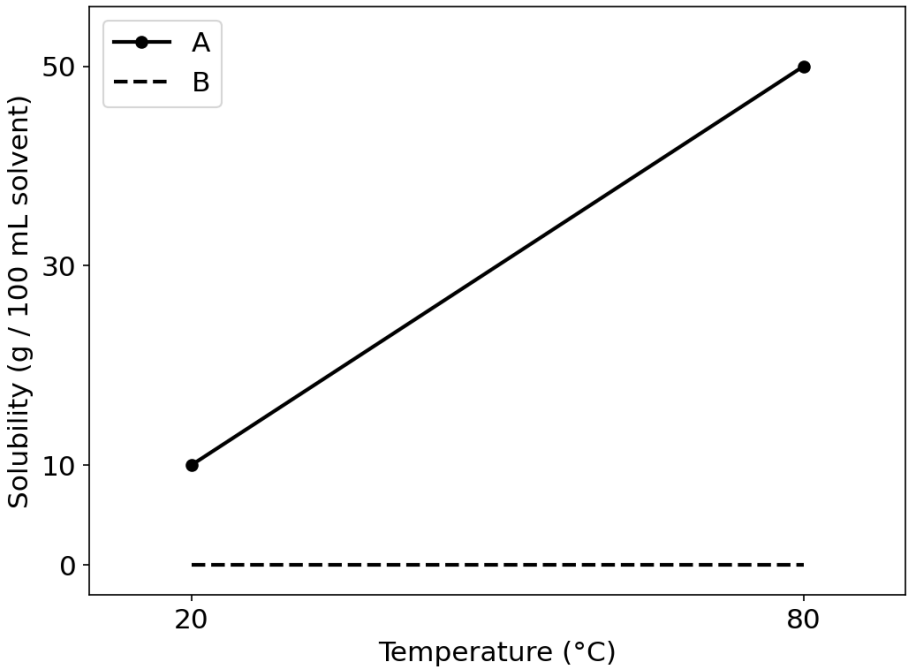
Q19 In an ideal one-electron ion an orbital has two angular nodes and one radial node, and its energy is -30.6 eV. Use E_n=-13.6 Z^2/n^2 eV and ignore fine structure. What are Z and the total number of spatial orbitals, across all subshells, degenerate with this orbital?
A) 6 and 16
B) 6 and 32
C) 3 and 16
D) 6 and 5

Q20 The E2+ ion has configuration [Ar]3d10. Taking E to be the neutral element and using the strict definition that a transition element has an incomplete d subshell in its atom or a common ion, which classification follows?
A) Period 3; transition element because the ion ends in d
B) Period 4; d-block element but not a transition element under this definition
C) Period 4; atomic number 28 because two electrons must be subtracted
D) Period 4; transition element because every d-block member qualifies

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Q21 A dry 100 mL mixture of CO and CO2 is measured at fixed temperature and pressure. Complete selective absorption of CO2 leaves 60 mL. For a separate original 100 mL sample, what are the initial CO volume fraction and the oxygen volume required for complete CO oxidation, with all gas volumes compared at the same conditions?
A) 0.40 and 20 mL
B) 0.60 and 60 mL
C) 0.40 and 30 mL
D) 0.60 and 30 mL

Q22 The graph gives solubility in grams per 100 mL of solvent: A dissolves to 50 g at 80 degrees C and 10 g at 20 degrees C; B is insoluble at both temperatures. A mixture of 30 g A and 5 g B is treated with 100 mL solvent at 80 degrees C. Which procedure and theoretical mass of pure A crystals are appropriate? Ignore solvent evaporation and mechanical losses.



A) Cool first, filter once, and collect 30 g pure A
B) Filter hot, cool the filtrate, and collect 30 g A
C) Filter hot, cool the filtrate, and collect 20 g A
D) Filter hot and collect the 5 g residue as pure A

Q23 Benzaldehyde, which has no alpha hydrogen, undergoes Cannizzaro reaction in concentrated base. Considering the original aldehyde carbon in the alcohol and carboxylate products, which oxidation states and classification are correct?
A) +1 and +3; oxidation of every reacting aldehyde
B) -1 and +3; disproportionation
C) -1 and +1; reduction without oxidation
D) -2 and +4; simple proton transfer

Q24 An alcohol C5H12O gives a ketone on controlled oxidation, and that ketone gives the iodoform test. Its usual more-substituted major constitutional alkene on acid dehydration has geometrical isomers. Ignore rearrangements and minor products. Which alcohol fits, and how many enantiomers does that alcohol have?
A) Pentan-3-ol; two
B) 3-Methylbutan-2-ol; two
C) 2-Methylbutan-2-ol; two
D) Pentan-2-ol; two

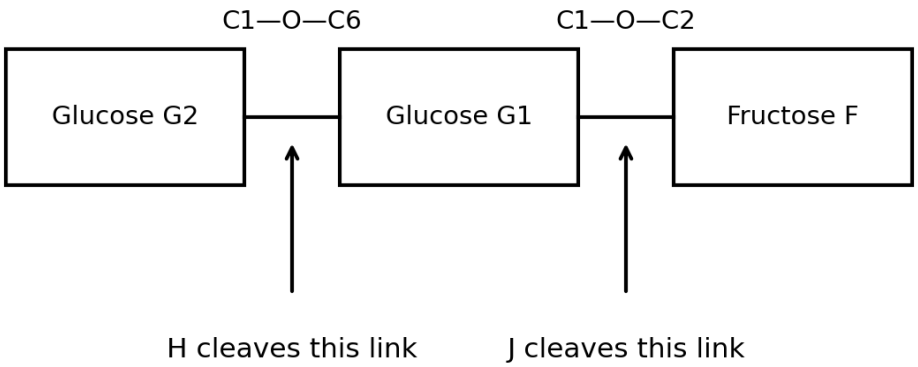
Q25 The supplied observations are that trimethylamine has greater gas-phase proton affinity than methylamine, but methylamine is the stronger base in water. Which explanation is consistent with this reversal?

A) Aqueous basicity also includes differential solvation of the protonated and neutral species
B) Trimethylamine loses its nitrogen lone pair permanently on entering water
C) Gas-phase proton affinity must always determine the aqueous order unchanged
D) Methylamine becomes a quaternary ammonium ion simply by dissolving

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Q26 The connectivity diagram specifies a trisaccharide: glucose G2 uses its anomeric C1 to bond to C6 of glucose G1; G1's anomeric C1 bonds to the anomeric C2 of fructose F. Enzyme H cleaves only the G2–G1 link, and enzyme J cleaves only the G1–F link. Each is applied separately to one initial molecule and cleavage is complete. After neutralisation, fructose counts as a reducing sugar in the usual alkaline test. What are the initial reducing character and the numbers of reducing product molecules after H and J, respectively?

Specified glycosidic links and selective cleavage



A) Nonreducing; two and one
B) Reducing; one and two
C) Nonreducing; one and two
D) Nonreducing; zero and one

Q27 A binary liquid has a minimum-boiling azeotrope at mole fraction xA=0.40 at pressure P1 and xA=0.60 at P2. Measurements on a liquid of xA=0.50 give vapour yA=0.46 at P1 and yA=0.55 at P2. For the first small vapour sample, which statement follows directly from these data?

A) It has the feed composition at both pressures because azeotropes cannot depend on pressure
B) It is enriched in A at P1 and depleted in A at P2
C) It is pure B at P1 and pure A at P2
D) It is depleted in A at P1 and enriched in A at P2

Q28 A hydrogen–oxygen fuel cell consumes hydrogen completely, has 100 percent Faradaic efficiency, and delivers a constant terminal voltage of 1.00 V. Formation of one mole liquid water releases 286 kJ enthalpy. Take F=96500 C mol−1 and neglect other energy stores. When 2F charge passes, what are the oxygen consumption and heat released to the surroundings at constant pressure?
A) 1.00 mol and 93 kJ
B) 0.500 mol and 93 kJ
C) 0.500 mol and 286 kJ
D) 1.00 mol and 193 kJ

Q29 A reaction has one constant rate law over the stated concentration range. The concentration falls from 0.80 to 0.40 M in 10 min, then from 0.40 to 0.20 M in a further 5 min. Among zero-, first- and second-order models, which model fits both intervals, and what concentration does it predict after 4 min for a fresh 0.30 M sample at the same temperature?
A) Zero order; 0.14 M
B) First order; 0.15 M
C) Second order; 0.10 M
D) Zero order; 0.26 M

Q30 An idealised sparingly soluble salt MA2 has Ksp=1.0×10−12, with M2+ and A− as its dissolution ions. A large buffer fixes [H+]=Ka=1.0×10−3 M for HA, so protonation of dissolved A− does not change the buffer pH. No A is initially present. Ignore metal hydrolysis, complexation and activity corrections. What is the molar solubility of MA2?
A) 5.0×10−7 M
B) 1.0×10−6 M
C) 1.0×10−4 M
D) 5.0×10−5 M

Mathematics

Q31 For which real k does $x^2 - y^2 - 2x + 2y + k = 0$ represent two intersecting straight lines, and where do those lines meet?
A) k=0; (0,0)
B) k=0; (1,1)
C) k=1; (1,1)
D) k=-1; (1,-1)

Q32 A circle passes through (1,0) and (0,2) and intersects the unit circle centred at the origin orthogonally. What are its centre and radius?
A) (1,5/4); 5/4
B) (1,5/4); √41/4
C) (1,3/4); 3/4
D) (0,1); 1

Q33 Let f(x)=x²+x on [0,1] and let g:[0,2]→[0,1] be its inverse. What is the area between y=f(x) and y=g(x) for 0≤x≤1?
A) (5√5-7)/12
B) 1/2
C) (17-5√5)/12
D) 5/6

Q34 Evaluate $\int_0^{\pi/2} \frac{dx}{1+\sin x}$.
A) π/2
B) 2
C) 1/2
D) 1

Q35 For y=x², consider its normals at points (t,t²) with t≠0. What is the set of their y-intercepts, and is its greatest lower bound attained by one of these normals?
A) [1/2,∞); attained
B) (1/2,∞); not attained
C) [0,∞); attained
D) All real numbers; no finite lower bound

Q36 The formula f(x)=x³-3x is restricted to domain [1,∞). Which is the smallest real codomain making f bijective, and what happens if that codomain is replaced by [0,∞) without changing the domain or formula?
A) [-2,∞); the proposed replacement is not a function into that codomain
B) [-2,∞); the replacement remains bijective
C) [0,∞); the replacement is automatically onto
D) The real line; the replacement is one-to-one but not onto

Q37 A real 2×2 matrix X satisfies $X + X^T + XX^T = 3I$, tr X=0 and det(I+X)>0. What is det X?
A) 3
B) 1
C) 0
D) -3

Q38 Independent binary events X,Y have probabilities 1/2 and 1/3. A selection event S has probabilities 1, 1/2, 1/4 and q conditional on (X,Y) being (1,1),(1,0),(0,1),(0,0), respectively. For what q are X and Y independent after conditioning on S, and then what is P(X|S)?
A) q=1/4; 2/3
B) q=1/8; 2/3
C) q=1/8; 4/5
D) q=0; 4/5

Q39 For distinct solutions x in [0,2π), consider $\sin^2 x - a \sin x = 0$. What are the solution counts at a=1, a=1/2 and a=3/2, respectively?
A) 2,4,2
B) 3,3,2
C) 4,4,2
D) 3,4,2

Q40 A plane has equation x/2+y/3+z/6=1. Let A be the area of its intercept triangle with the positive coordinate axes and d its distance from the origin. What is Ad?
A) 18
B) 6
C) 3√14
D) 36

Q41 Unit vectors a,b are perpendicular. A vector c has |c|=3, c·a=1 and c·b=2. Its component along a×b is negative. What is $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$?
A) 2
B) -√5
C) -2
D) -3

Q42 Let z range over the complex line Re z=2 and set w=1/z=u+iv. What is the exact locus of w?
A) (u-1/2)²+v²=1/4, including 0
B) (u-1/4)²+v²=1/16, including 0
C) The line u=1/2
D) (u-1/4)²+v²=1/16, excluding 0

Q43 For which real a is $x^2 - 2ax + 2a$ strictly positive at every x in the closed interval [0,1]?
A) 0<a<2
B) a≥0
C) a>0
D) a>2

Q44 Evaluate $\sum_{k=0}^{8} (-1)^k k(k-1)(k-2) \binom{8}{k} 2^{8-k}$.
A) 336
B) -336
C) 0
D) -168

Q45 A real geometric progression has seven nonzero terms, and its fourth term is -2. What is the product of all seven terms?
A) 128
B) -64
C) 64
D) -128

Physics

Q46 A camera moves horizontally at 2 m s−1 in the laboratory and is alongside a projectile at launch, with its axes remaining parallel to laboratory axes. Relative to the camera the projectile is at (6,15) m after 1 s and (12,20) m after 2 s. Air resistance is absent and g=10 m s−2. What are the laboratory launch angle above horizontal and the camera-relative horizontal coordinate when the projectile next returns to its 1-second height?
A) arctan(5/2); 18 m
B) arctan(10/3); 18 m
C) arctan(5/2); 24 m
D) arctan(5/3); 12 m

Q47 A 1 kg particle enters a smooth stationary U-shaped guide with velocity +2 m s−1 along x and leaves with velocity -2 m s−1 along x. The guide's force is always perpendicular to instantaneous velocity and gravity is absent. What are the work by the guide and its total x impulse on the particle?
A) 0 J and -4 N s
B) 4 J and 0 N s
C) 0 J and 0 N s
D) -4 J and -4 N s

Q48 A uniform slender rod of mass M and length L is initially at rest on a frictionless horizontal surface. A horizontal impulse J perpendicular to the rod is applied at a point L/6 from its centre. What fraction of the resulting total kinetic energy is rotational about the centre of mass? Use $I_{CM}=ML^2/12$.
A) 1/2
B) 3/4
C) 1/4
D) 1/3

Q49 A clean capillary tube with fixed radius and zero contact angle gives a 4 cm rise of a liquid on Earth. The same tube and liquid are placed where gravity is g/6, with surface tension, density and contact angle unchanged and enough liquid available. Which rise and pressure jump across the meniscus follow?
A) 24 cm; six times the Earth pressure jump
B) 24 cm; the same pressure jump as on Earth
C) 4 cm; the same pressure jump as on Earth
D) 2/3 cm; one sixth the pressure jump

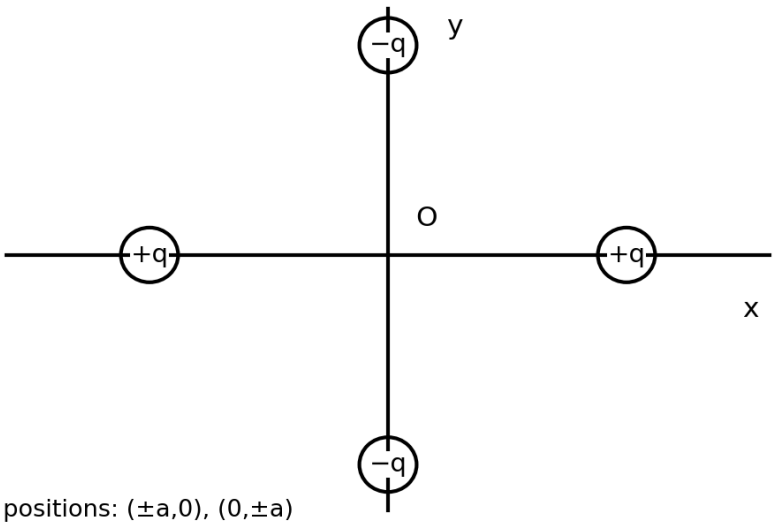
Q50 A reversible engine works between 600 K and a 300 K room, absorbing 1200 J per interval. All its work drives a reversible refrigerator between that room and a 250 K cold chamber. During the same interval, 400 J also leaks from the room into the cold chamber. What are the net heat removed from the chamber and the net heat gained by the room?
A) 3000 J and 4200 J
B) 2600 J and 4200 J
C) 2200 J and 3400 J
D) 2600 J and 3800 J

Q51 Two ideal string segments, each length L, have linear densities μ and 4μ and the same tension T. Their junction and both outer ends are nodes of a normal mode. What is the lowest possible common frequency, and how many distinct nodes occur across the whole string, counting the outer ends and the junction?
A) √(T/μ)/(4L); three
B) √(T/μ)/L; five
C) √(T/μ)/(2L); four
D) √(T/μ)/(2L); three

Q52 A uniform solid sphere has mass M and radius R. A concentric thin spherical shell of mass M has radius 2R. Where is the magnitude of the total gravitational field largest, and what is that maximum?
A) At 2R just outside; GM/(2R²)
B) At R; GM/R²
C) At the centre; 2GM/R²
D) At R; 2GM/R²

Q53 Four fixed charges lie in the xy-plane: +q at (±a,0) and -q at (0,±a). A positive test charge is constrained in turn to move near the origin along each coordinate axis. Which local equilibrium classification is correct?

Four fixed charges; test point at the origin



Charge positions: (±a,0), (0,±a)

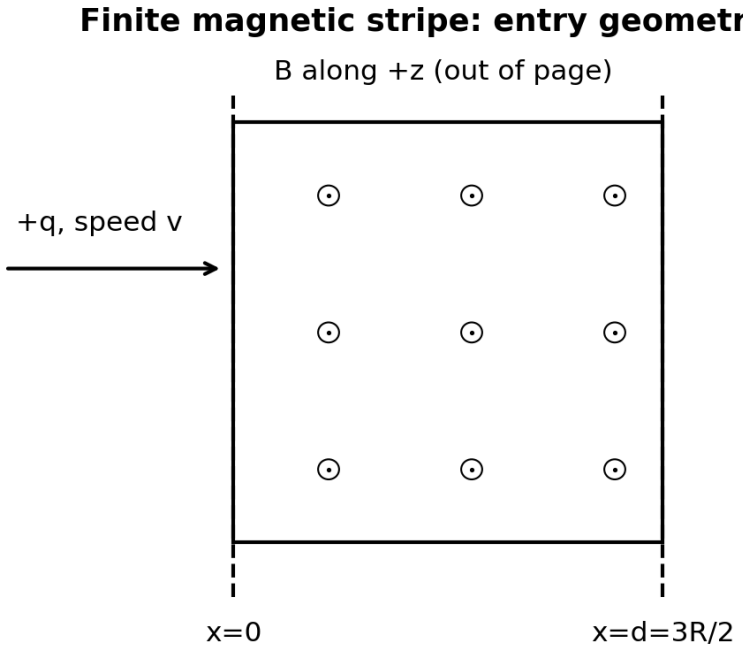
A) Restoring along x, unstable along y, neutral along z
B) Unstable along x, restoring along y, neutral along z
C) Restoring along every axis because potential is zero
D) Neutral along every axis because the field at the origin is zero

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Q54 Three equal isolated capacitors A,B,C each have capacitance C0. Initially A has voltage V and B,C are uncharged. First A and B are connected positive plate to positive plate through a resistor until equilibrium. They are disconnected, leaving B isolated. Then A is connected similarly to C until equilibrium. What are B's final voltage and the total heat dissipated in the two resistors?
A) V/3; $C0V^2/3$
B) V/4; $5C0V^2/16$
C) V/2; $C0V^2/4$
D) V/2; $5C0V^2/16$

Q55 Cells of emf 12 V and 6 V, each with internal resistance 1 ohm, are connected in parallel with their positive terminals joined and a load R across the common terminals. What load resistance marks zero current in the 6 V cell, and what charging current enters that cell when R=2 ohm?
A) 1 ohm; 6/5 A
B) 2 ohm; 6/5 A
C) 1 ohm; 18/5 A
D) 1/2 ohm; 6 A

Q56 A positive particle enters at x=0 with velocity v along +x into a uniform magnetic field +B along z confined to 0<x<d. Let R=mv/(qB) and take d=3R/2. With no other forces, through which boundary does it leave, after what time, and in what direction?



A) Right boundary; πm/(2qB); -y
B) Left boundary; πm/(qB); -x
C) Right boundary; πm/(qB); +x
D) Left boundary; 2πm/(qB); +x

Q57 A series RLC circuit is driven by a sinusoidal source of fixed voltage amplitude E. L=1 H, C=0.010 F and R=10 ohm. At what angular frequency is the capacitor voltage amplitude greatest, and what is that greatest amplitude divided by E?
A) 10 rad s−1; 1
B) 10 rad s−1; √2
C) 5√2 rad s−1; 2/√3
D) 5 rad s−1; 2

Q58 A compound microscope has objective focal length 1 cm, eyepiece focal length 5 cm and lens separation 11 cm. Initially the final image is at infinity. Keeping both lenses fixed, the object is moved until the eyepiece forms a virtual final image 25 cm from it on its incident side. What is the new object distance from the objective, and how far and in which direction was it moved?
A) 6/5 cm; 1/35 cm away
B) 41/35 cm; 1/35 cm toward
C) 41/35 cm; 1/35 cm away
D) 35/41 cm; 1/5 cm toward

Q59 Two equal plane electromagnetic waves in vacuum have electric fields $E_0 \cos(kx - \omega t)\hat{y}$ and $E_0 \cos(kx + \omega t)\hat{y}$. Their magnetic fields point along +z and -z, respectively, for positive values of the corresponding cosine. At x=0, which total fields and comparison with the single-travelling-wave relation E=cB are correct?
A) E=0 and B=2E0 cos(ωt)/c; magnetic fields add
B) E=2E0 cos(ωt) along y and B=E/c along z; the single-wave ratio remains valid
C) Both fields vanish at all times; no electric field remains
D) E=2E0 cos(ωt) along y and B=0; their sum generally does not obey the single-wave ratio

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Q60 Neutral-atom masses of a parent (A,Z) and daughter (A,Z-1) differ by 1.5 electron masses, the parent being heavier. Neglect electronic binding, nuclear excitation and neutrino mass. What is the corresponding bare-nucleus mass difference, and is spontaneous positron emission energetically possible?
A) 1.5 me; possible
B) 2.5 me; possible
C) 0.5 me; impossible
D) 0.5 me; possible

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A Four protist groups separated by coverings, flagella and a feeding-to-spore transition

Answer: (A) Euglenoid, diatom, dinoflagellate and slime mould.

Why: P combines a flexible covering with condition-dependent nutritional modes typical of euglenoids. Q's two-part siliceous wall is diagnostic of diatoms, whereas R's plates and flagellar arrangement fit dinoflagellates. S's feeding plasmodial mass and subsequent spore-bearing structures distinguish the slime-mould life form. All four observations must agree with the assignment; photosynthesis alone would not separate the first three groups.

Why not the others: (B) swaps the pellicle and siliceous-wall organisms and assigns the plasmodial stage to dinoflagellates. (C) gives the silica wall to a slime mould and places the flagellar plates in a diatom. (D) assigns a flexible photosynthetic cell to the plasmodial group and reverses the other diagnostic coverings.

Key point: Use covering, movement and life-cycle evidence together instead of classifying a protist by nutrition alone.

Q2 / B Nodes and buds identify an underground stem despite a root-like storage role

Answer: (B) P is a modified stem and Q a modified root.

Why: Nodes, scale leaves and axillary buds belong to the shoot organisation and identify P as a modified stem despite its underground position. Q lacks those features and has the stated root organisation. Storage is a function that can be performed by different organs, so neither the shared function nor location overrides the morphological evidence.

Why not the others: (A) treats underground position as a sufficient root character. (C) reverses the significance of nodes and axillary buds. (D) confuses the chemical contents of an organ with its developmental identity.

Key point: Determine organ identity from its structural markers before using its location or function.

Q3 / C Chromatin repeat length distinguishes core histone count from linker DNA

Answer: (C) 240 core histone molecules and 1620 linker bp.

Given: Total DNA 6000 bp; repeat 200 bp; eight core histones and 146 wrapped bp per repeat.

Formula/Principle: Count repeats first, then distinguish core protein count from unwrapped DNA amount.

Step 1:

$$N_{\text{repeat}} = 6000/200 = 30.$$

Step 2:

$$N_{\text{histone}} = 30(8) = 240 \text{ molecules}.$$

Step 3: Linker length per repeat is $200 - 146 = 54$ bp, so

$$L_{\text{linker}} = 30(54) = 1620 \text{ bp}.$$

Why: The core is an octamer, not one histone molecule, while a repeat includes both wrapped and linker DNA. The two requested quantities therefore use different multipliers after the same repeat count. Packaging the DNA does not cut away the linker or change the base-pair total.

Why not the others: (A) counts cores instead of histone molecules. (B) reports the wrapped DNA length rather than the linker length. (D) doubles both counts as though a base pair were two independent repeats.

Key point: Keep structural-unit count, protein stoichiometry and DNA length in separate accounts.

Q4 / A Regulator-dependent proliferation and organ formation separate dedifferentiation from redifferentiation

Answer: (A) Dedifferentiation followed by redifferentiation.

Why: Previously differentiated cells regaining proliferative capacity illustrate dedifferentiation. Their descendants then acquiring organised shoot characteristics illustrate redifferentiation. Changing the hormonal environment can regulate which existing developmental capacities are expressed without requiring a new DNA sequence for every tissue type. The experiment therefore separates a change of cellular state from a required mutational explanation.

Why not the others: (D) introduces a reduction division and gametic stage not observed in the tissue culture. (B) contradicts the later formation of organised shoots. (C) treats a regulated developmental response as proof of mutations despite the stated sequence evidence.

Key point: A cell can change developmental state and fate through regulation without acquiring a new genome.

Q5 / D Preserved effectors distinguish movement coordination from central respiratory-rhythm control

Answer: (D) Cerebellum and medulla oblongata.

Why: The cerebellum contributes to coordination, precision and balance, so impaired coordination need not mean the muscles themselves cannot produce force. The medulla contains important centres controlling respiratory rhythm; muscle contractility does not substitute for that coordinated neural drive. Preserved breathing in X and preserved direct muscle responses in Y help separate coordination and rhythm control from a general failure of all motor tissue. These are idealised functional assignments, not a unique clinical diagnosis from two symptoms.

Why not the others: (B) exchanges the main coordination and respiratory-rhythm assignments. (C) does not match the stated coordination role in X and confuses loss of respiratory drive with a necessary loss of muscle contractility. (A) substitutes peripheral sensory interruption for the given central coordination comparison.

Key point: Distinguish the ability of an effector to respond from the central organisation of its activity.

Q6 / B Anther nutrition failure and exine chemical resistance provide independent pollen evidence

Answer: (B) Tapetum and sporopollenin.

Why: The tapetum is the nutritive anther layer associated with developing pollen, so damage before pollen maturation can impair pollen regardless of which compatible stigma later receives it. Sporopollenin accounts for the remarkable resistance of the exine. The stigma-transfer result places the defect before a particular pollen-stigma compatibility interaction, while the chemical-resistance result identifies a different structural feature of pollen.

Why not the others: (A) locates the original treatment in a receptive female surface rather than the stated anther layer and replaces the resistant exine polymer by a different wall component. (C) concerns an ovule tissue and a non-exine wall material. (D) assigns a post-fertilisation nutritive tissue to anther development and substitutes lignin for sporopollenin.

Key point: Use the developmental location of a defect separately from the chemical identity of a protective wall.

Q7 / C Maternal centromeric marker identity separates meiosis-I from meiosis-II nondisjunction

Answer: (C) Maternal meiosis I and maternal meiosis II.

Why: Failure to separate homologous maternal X chromosomes in meiosis I can retain both different centromeric markers in one egg. After a normal meiosis-I separation, failure of sister chromatids to separate in meiosis II can instead give an egg with two copies bearing the same marker. Fertilisation by a Y-bearing sperm gives the stated XXY complements. The absence of paternal XB from both children, together with the stated exclusions, localises the extra X material to the maternal route. The centromere/no-recombination condition is necessary because distal recombination could otherwise obscure the sister-versus-homologue marker distinction.

Why not the others: (A) reverses retention of different homologues and duplicate sisters. (B) cannot explain the identical marker pair in the second child under the given assumptions. (D) cannot retain both different maternal centromeric markers in the first child after normal homologue separation.

Key point: Centromere-linked heterozygosity versus duplicate sister markers can distinguish which meiotic separation failed.

Q8 / D Two fungal industrial products distinguished by enzyme inhibition versus immunosuppression

Answer: (D) Statin and cyclosporin A.

Why: Monascus-derived statins are associated with inhibition of cholesterol synthesis, whereas cyclosporin A from Trichoderma polysporum is used for its immunosuppressive action. The target evidence distinguishes these products from antibacterial agents and clot-dissolving enzymes even though all may be discussed among useful microbial products. Being produced by a fungus does not make every product an antibiotic or give it the same cellular target.

Why not the others: (A) replaces the cholesterol-synthesis inhibitor by an antibacterial drug and the immunosuppressant by a clot-dissolving enzyme. (B) reverses the two target-specific assignments. (C) misassigns both the clot-dissolving and antibacterial products to the stated activities.

Key point: Identify a microbial product by its source and biological target rather than assuming a shared source implies a shared action.

Q9 / A Limited corrected-cell persistence motivates a progenitor-level gene-therapy strategy

Answer: (A) A persistent corrected progenitor population.

Why: Enzyme production in the returned lymphocytes can be functional yet temporary because those cells do not persist indefinitely. A suitably corrected self-renewing or persistent source of descendants addresses that cellular limitation rather than merely increasing one transient enzyme pulse. The conclusion is a mechanism within the stated model, not a claim that every clinical gene-therapy procedure automatically cures the disorder.

Why not the others: (B) removes the genetic instruction responsible for restored enzyme production. (C) supplies cells that still lack the corrected function and gives an unsupported lifespan comparison. (D) assumes secreted or cellular enzyme activity rewrites unrelated genomes, which does not follow.

Key point: Distinguish successful gene expression in a short-lived cell from persistence of a corrected cell lineage.

Q10 / B A seed-level cost coexists with a measured mutualistic partner-level outcome

Answer: (B) A net mutualistic interaction can include individual seed costs.

Why: The measured increase in viable-seed production establishes a reproductive benefit to the tree under the comparison, while the wasp gains a required reproductive opportunity. Consumption of some seeds is a real local cost, but it does not erase the larger measured net benefit. Interaction labels therefore depend on the partner-level outcomes being assessed; a mutualistic relationship need not mean that every component of each organism is unharmed.

Why not the others: (A) treats one local cost as sufficient to reverse both measured net outcomes. (C) assumes one partner's reproduction excludes benefit to the other. (D) wrongly excludes the resource exchanges that often enable mutualism.

Key point: Evaluate net partner outcomes at the stated scale before naming a population interaction.

Q11 / C Globin-chain amount distinguished from a structural variant's oxygen-dependent effect

Answer: (C) Quantitative thalassaemia versus qualitative sickle-cell change.

Why: A reduction in the amount of a globin-chain type changes the balance of haemoglobin-chain production, the quantitative distinction emphasised for thalassaemia. The stated structural substitution and oxygen-dependent sickling in Q fit a qualitative haemoglobin defect. The absence of nuclei in mature red cells does not prevent inherited instructions in their precursor lineage from determining which proteins those cells contain. Amount and molecular structure are therefore separate dimensions of an inherited protein disorder.

Why not the others: (A) reverses quantitative and qualitative effects. (B) ignores the stated inherited globin-production and sequence mechanisms. (D) confuses the mature cell's structure with the genome of the organism and its blood-cell precursors.

Key point: Ask whether an inherited protein defect changes how much is made or the properties of the molecule made.

Q12 / D A nonmonotone hominin specimen series is interpreted separately from homology and functional analogy

Answer: (D) No strict sample-volume increase, and skeletal correspondence supports homology.

Why: The Neanderthal-to-later-sapiens comparison decreases from 1400 to 1350 cm3 in the stated measurements, so a strictly increasing volume rule fails for these samples. Relative dates and one measurement do not by themselves establish a direct ancestral chain or a universal progress ladder. In the anatomical comparison, correspondence of underlying forelimb elements supports common structural ancestry despite changed use. Flight alone supplies a functional resemblance between bird and insect wings, not the same evidence of structural homology. The volumes are supplied specimen observations, not claims that every individual in a named group had one fixed cranial capacity.

Why not the others: (A) contradicts the final measured decrease and equates a shared function with recent ancestry. (B) infers a lineage directly from three measurements and mistakes divergent function for absence of homology. (C) uses brain magnitude to reverse or invent ancestry and substitutes functional analogy for construction-based evidence.

Key point: Keep chronological evidence, a measured trait and homologous structural correspondence as distinct lines of evolutionary reasoning.

Q13 / A Multiaxial joint geometry and urate evidence distinguish a joint disorder from muscle failure

Answer: (A) Ball-and-socket articulation and gout-related inflammation.

Why: A ball-and-socket joint permits the stated multiaxial movement and rotation, unlike a simple hinge. Uric-acid crystal deposition with joint inflammation is the relevant mechanism for gout in the comparison. Preserved force after direct stimulation also separates joint-limited movement from a necessary intrinsic failure of all contractile machinery. These findings address different levels: articulation geometry and an inflammatory limitation of movement.

Why not the others: (B) limits the movement to a hinge pattern and substitutes decreased bone mass for the crystal evidence. (C) assigns a limited gliding joint to the shoulder-like movement and a different autoimmune target to the inflammation. (D) contradicts both the observed joint mobility and the retained muscle force.

Key point: Distinguish joint geometry and joint inflammation from defects of bone amount or muscle activation.

Q14 / D Intact lipid redistribution distinguished from carbohydrate and peptide hydrolysis

Answer: (D) Redistribution of intact phospholipid molecules.

Why: The detergent result changes how existing lipid molecules associate, while the stem explicitly excludes new covalent species. The enzyme results instead include smaller covalently connected products produced by hydrolysing glycosidic or peptide linkages. Catalysts can accelerate bond-changing reactions without being permanently consumed; their catalytic status does not make those reactions physical mixing.

Why not the others: (A) wrongly calls peptide bonds noncovalent. (B) denies starch's polymeric carbohydrate nature and ignores the stated hydrolysis products. (C) confuses catalyst conservation with conservation of every substrate bond.

Key point: Use evidence for changed covalent species to distinguish chemical breakdown from altered aggregation.

Q15 / B Cereal embryo geometry separates two protective sheaths from its lateral cotyledon

Answer: (B) Coleoptile, coleorhiza and scutellum.

Why: In the cereal embryo, the shoot-associated plumule is enclosed by the coleoptile, while the radicle is enclosed by the coleorhiza. The single lateral cotyledon is the scutellum, distinct from the surrounding nutritive endosperm. Reading the sheaths relative to the two embryonal-axis ends prevents exchanging their names simply because both are protective structures.

Why not the others: (A) swaps the two sheaths and calls the lateral embryonic cotyledon endosperm. (C) places a cotyledon and storage tissue where the two protective sheaths are shown. (D) again replaces axis-associated embryonic structures with the wrong tissue identities.

Key point: Identify cereal-embryo structures by their position and developmental identity, not merely by a shared protective or storage role.

Chemistry

Q16 / B Lewis-adduct geometry distinguished from oxidation-state change

Answer: (B) Boron becomes tetrahedral without a redox change.

Step 1: BF3 has three bonding regions and an incomplete octet at boron. NH3 supplies an electron pair to a new B–N bond, giving four bonding regions. The electron-pair arrangement becomes tetrahedral.

Step 2: Coordinate bonding assigns the origin of a shared pair; it is not itself electron transfer for oxidation-state bookkeeping. Fluorine remains -1, hydrogen +1, boron +3 and nitrogen -3.

Why: Lewis acid–base association changes coordination and geometry without requiring oxidation or reduction.

Why not the others: (A) Four bond directions do not give linear geometry. (C) The oxidation number of boron does not decrease. (D) The starting boron geometry and the proposed proton-transfer mechanism are both wrong.

Key point: Coordination number and oxidation number describe different properties.

Q17 / D Total-metal versus selective EDTA assay identifiability

Answer: (D) The first assay gives a sum; the second resolves magnesium as 0.400 mmol.

Given: Both assays use the same sample amount and 1:1 metal–EDTA stoichiometry.

Step 1: The total assay supplies one equation,

$$n_{Ca} + n_{Mg} = 1.00 \text{ mmol}.$$

Infinitely many nonnegative pairs satisfy it.

Step 2: Selectivity adds $n_{Ca} = 0.600$ mmol. Hence

$$n_{Mg} = 1.00 - 0.600 = 0.400 \text{ mmol}.$$

Why: EDTA's denticity counts donor atoms attached to one metal, not the number of metal ions consumed per ligand.

Why not the others: (A) A sum does not imply equality. (B) The total assay provides no ratio. (C) Six donor atoms do not imply six metal ions per EDTA.

Key point: Independent selective measurements are needed to separate components of a total.

Q18 / C Compare manganate and permanganate magnetic counts without testing beyond-scope colour mechanism

Answer: (C) The spin-only moments are √3 μB and zero, and colour alone does not establish an unpaired electron.

Given: Neutral Mn is [Ar]3d5 4s2.

Formula/Principle: Oxidation-state bookkeeping gives formal d counts; spin-only moment depends on unpaired electrons.

Step 1: Charge balance gives Mn +6 in manganate and +7 in permanganate. Their formal configurations are d1 and d0.

Step 2: The unpaired counts are therefore one and zero.

$$\mu_{VI} = \sqrt{1(1+2)}\,\mu_B = \sqrt{3}\,\mu_B, \qquad \mu_{VII} = 0.$$

Why: The coloured permanganate is a counterexample to the claim that every coloured species must have an unpaired ground-state electron. No detailed mechanism of its colour is required.

Why not the others: (A) assigns an unpaired electron to d0. (B) reverses the electron-count trend on oxidation. (D) ignores the single d electron of manganate.

Key point: Infer magnetic moment from electron count rather than colour.

Q19 / A Infer hydrogenic nuclear charge and full shell degeneracy from nodes

Answer: (A) Z=6 and there are 16 spatial orbitals.

Given: Angular nodes 2; radial nodes 1; energy -30.6 eV.

Formula/Principle: Angular nodes equal l; radial nodes equal n-l-1. In the ideal hydrogenic model energy depends on n only.

Step 1: $l = 2$ and $n - 2 - 1 = 1$ give $n = 4$.

Step 2:

$$-30.6 = -13.6Z^2/16 \;\Rightarrow\; Z^2 = 36, \quad Z = 6.$$

Step 3: All permitted $l = 0, 1, 2, 3$ share this energy. Their spatial-orbital counts are

$$1 + 3 + 5 + 7 = 16.$$

Why: Node information identifies a particular subshell, but hydrogenic degeneracy includes every subshell with the same n.

Why not the others: (B) 32 counts spin states, not spatial orbitals. (C) Z=3 gives a different energy at n=4. (D) Five counts only the d orbitals, omitting other degenerate subshells.

Key point: Subshell identification does not restrict the energy degeneracy to that subshell.

Q20 / B Recover neutral zinc period and distinguish d block from transition definition

Answer: (B) E is zinc, a period-4 d-block element excluded by the stated transition-element definition.

Step 1: The ion has $18 + 10 = 28$ electrons. Neutral E has $28 + 2 = 30$, with configuration [Ar]3d10 4s2.

Step 2: Its highest occupied principal shell is n=4. The atom and its common +2 ion both retain a full d subshell.

Why: Block membership and the narrower transition-element definition need not coincide. The lost 4s electrons must be restored before locating the neutral atom's period.

Why not the others: (A) The ion's outer occupied shell does not locate the neutral atom's period. (C) Forming a cation removes electrons; reconstructing its atom adds them. (D) It ignores the explicit incomplete-d condition.

Key point: Classify the neutral element after restoring ionic charge.

Q21 / D Selective gas absorption followed by oxidation demand

Answer: (D) The CO fraction is 0.60 and oxygen consumption is 30 mL.

Given: Unabsorbed gas is CO; volume comparisons share temperature and pressure.

Formula/Principle: Gas volumes at common conditions follow balanced mole ratios.

Step 1: The unabsorbed gas is CO. Its volume fraction is

$$x_{CO} = 60/100 = 0.60.$$

Step 2: Balance oxidation:

$$2\mathrm{CO} + \mathrm{O_2} \longrightarrow 2\mathrm{CO_2}.$$

At common temperature and pressure volume ratios equal mole ratios, so

$$V_{O_2} = 60/2 = 30 \text{ mL}.$$

Why: Absorption identifies composition; balanced reaction stoichiometry then determines oxygen demand. Initial CO2 is already fully oxidised.

Why not the others: (A) It treats the absorbed volume as CO. (B) It uses a 1:1 CO:O2 ratio. (C) Its oxygen value fits CO=60 mL but its stated fraction contradicts that amount.

Key point: Identify the retained gas before applying the reaction ratio.

Q22 / C Hot filtration and finite recrystallisation recovery

Answer: (C) Hot filtration removes B; cooling gives 20 g A.

Step 1: Hot solvent can dissolve 50 g A, exceeding the 30 g present. Thus all A dissolves while 5 g B remains solid. Hot filtration removes B without deliberately crystallising A onto it.

Step 2: At 20 degrees C the mother liquor retains 10 g A. The recovered crystal mass is

$$m_{A,crystal} = 30 - 10 = 20 \text{ g}.$$

Why: Recrystallisation exploits the temperature dependence of solubility, whereas hot filtration removes the temperature-insoluble impurity.

Why not the others: (A) Cooling before removing B mixes B with the new A crystals. (B) It neglects dissolved A left in the cold mother liquor. (D) The hot insoluble residue is B.

Key point: Purification and theoretical recovery require separate impurity and solubility accounting.

Q23 / B Cannizzaro reaction tracked by carbon oxidation-state balance

Answer: (B) The carbon reaches -1 in benzyl alcohol and +3 in benzoate: disproportionation.

Step 1: The net reaction is

$$2C_6H_5CHO + OH^- \longrightarrow C_6H_5CH_2OH + C_6H_5COO^-.$$

Step 2: At the aldehyde carbon, C=O contributes +2, C–H contributes -1, and C–C contributes 0, giving +1. In the alcohol, C–O and two C–H bonds give $+1 - 2 = -1$. In carboxylate, oxygen bonds give +3.

Step 3: One carbon gains two electrons formally and the other loses two; the changes balance.

Why: The same starting species is both oxidised and reduced. Lack of alpha hydrogen prevents the ordinary self-aldol route.

Why not the others: (A) The alcohol carbon is reduced. (C) It assigns the carboxylate its starting oxidation state. (D) Neither product carbon has the listed state, and redox has occurred.

Key point: Follow the reacting carbon, not the oxidation state of the whole molecule.

Q24 / D Intersect alcohol oxidation, iodoform and alkene stereochemical constraints

Answer: (D) Pentan-2-ol fits all three observations and has two enantiomers.

Step 1: Ketone formation selects a secondary alcohol. The tertiary option is excluded under controlled oxidation without carbon-skeleton cleavage.

Step 2: Iodoform-positive oxidation requires a methyl ketone, hence an initial CH3–CH(OH)– group. Pentan-3-ol instead gives pentan-3-one and fails this test.

Step 3: 3-Methylbutan-2-ol can give 2-methylbut-2-ene, but one alkene carbon has two methyl groups, preventing E/Z isomerism. Pentan-2-ol gives pent-2-ene, which has two different substituents at each double-bond carbon.

Step 4: Its alcohol carbon bears H, OH, methyl and propyl, four different groups. One stereocentre gives two enantiomers.

Why: Each experiment removes a different structural ambiguity; no single test suffices.

Why not the others: (A) Its ketone is not a methyl ketone. (B) Its stated major alkene lacks geometrical isomers. (C) A tertiary alcohol does not give the stipulated ketone and its two methyl groups prevent the claimed stereocentre.

Key point: Apply oxidation, functional-group testing and stereochemistry to the same candidate.

Q25 / A Reconcile gas-phase and aqueous amine basicity orders

Answer: (A) Solvation changes the free-energy balance for protonation.

Step 1: Gas-phase proton affinity compares proton binding without a solvent. Electron donation by alkyl groups contributes to that comparison.

Step 2: In water the equilibrium compares solvated bases and solvated conjugate acids. Hydrogen bonding and the accessibility of the protonated nitrogen region affect their relative stabilisation. These solvent contributions can outweigh the gas-phase trend in the stated pair.

Why: Basicity is an equilibrium property of a specified medium, not a ranking transferable without its free-energy environment.

Why not the others: (B) Solvation does not permanently remove a lone pair from neutral trimethylamine. (C) It contradicts the supplied reversal and omits solvent contributions. (D) Protonation gives an ammonium ion with an N–H bond, not automatic carbon alkylation to a quaternary ion.

Key point: Compare bases in the same solvent and account for conjugate-acid solvation.

Q26 / C Link-specific partial hydrolysis and reducing-product counts

Answer: (C) The initial molecule is nonreducing; H gives one reducing product and J gives two.

Step 1: All three anomeric centres are involved in glycosidic bonds: G2 C1, G1 C1 and F C2. No free anomeric group remains, so the initial molecule is nonreducing.

Step 2: H releases free glucose G2, which is reducing. The remaining G1–F unit still links both its anomeric centres and is nonreducing. The count is one.

Step 3: J releases free fructose, reducing under the stipulated test. The remaining G2–G1 disaccharide has G1 C1 free, so it too is reducing. The count is two.

Why: Reducing character depends on an available anomeric group and the test conditions, not merely the number of glycosidic links.

Why not the others: (A) It reverses the two cleavage outcomes. (B) It overlooks that the initial G1 C1 is bonded to F C2. (D) It fails to recognise the free glucose released by H and one of the two reducing products from J.

Key point: Track the specific anomeric carbon freed by each cleavage.

Q27 / D Pressure-specific azeotrope data reverse the first-vapour enrichment direction

Answer: (D) Relative to the 0.50 feed, 0.46 is depleted and 0.55 is enriched.

Step 1: At P1,

$$y_A - x_A = 0.46 - 0.50 = -0.04.$$

Thus the first vapour contains a smaller mole fraction of A than the liquid.

Step 2: At P2,

$$y_A - x_A = 0.55 - 0.50 = +0.05.$$

The direction reverses. At neither pressure is the given feed itself at its listed azeotropic composition.

Why: The phase compositions are measured at the specified pressure. A constant-boiling composition at one pressure is not a universal composition fixed for every pressure.

Why not the others: (A) It contradicts both measured vapour compositions and the stated pressure-dependent azeotropes. (B) It reverses the signs of both differences. (C) Neither 0.46 nor 0.55 is a pure-component mole fraction.

Key point: Compare vapour with its own liquid at the same pressure before inferring enrichment.

Q28 / B Fuel-cell charge stoichiometry and measured-voltage heat budget

Answer: (B) Oxygen consumption is 0.500 mol and heat release is 93 kJ.

Given: Charge 2F passes at terminal voltage 1.00 V.

Formula/Principle: Electrical work is QV; reaction enthalpy supplies electrical work plus heat.

Step 1:

$$\mathrm{H_2} + \tfrac{1}{2}\mathrm{O_2} \longrightarrow \mathrm{H_2O\,(l)}$$

transfers two moles of electrons per mole of hydrogen. Charge 2F therefore consumes 1 mol H2 and 0.5 mol O2.

Step 2: Delivered electrical work is

$$W_{out} = QV = (2 \times 96500)(1.00) = 193000\ \mathrm{J} = 193\ \mathrm{kJ}.$$

Step 3: The released enthalpy is partitioned into electrical work and heat:

$$Q_{out} = 286 - 193 = 93\ \mathrm{kJ}.$$

Why: Reaction enthalpy is not identical to terminal electrical output. Both must be included in the constant-pressure energy budget.

Why not the others: (A) It doubles the oxygen demand. (C) It ignores the energy delivered electrically. (D) It both doubles oxygen and calls the electrical output heat.

Key point: Use electron stoichiometry before splitting chemical energy into work and heat.

Q29 / A Order identification from successive unequal halving intervals and fresh-sample prediction

Answer: (A) Constant absolute consumption gives zero order and 0.14 M.

Given: Successive halving intervals are 10 and 5 min.

Formula/Principle: Zero-order concentration decreases linearly; first-order half-life is concentration independent.

Step 1: Interval rates are

$$(0.80 - 0.40)/10 = 0.040, \qquad (0.40 - 0.20)/5 = 0.040\ \mathrm{M\,min^{-1}}.$$

Both agree with concentration-independent rate. First-order halving times would be equal; second-order halving times would increase as concentration falls.

Step 2: For the fresh sample,

$$[A]_t = [A]_0 - kt = 0.30 - 0.040(4) = 0.14\ \mathrm{M}.$$

Why: Comparing successive halving intervals identifies how rate changes with concentration before extrapolating another initial condition.

Why not the others: (B) First order contradicts the unequal halving times. (C) Second order predicts the opposite halving-time trend. (D) It subtracts only 0.04 M rather than the four-minute consumption of 0.16 M.

Key point: Equal amounts per time and equal fractions per time imply different orders.

Q30 / C Buffered protonation coupled to a two-anion solubility product

Answer: (C) The solubility is 1.0×10−4 M.

Given: A fixed buffer has [H+]=Ka and dissolved salt supplies two anions per metal.

Formula/Principle: Combine acid dissociation, anion balance and the free-ion solubility product.

Step 1: $K_a = [H^+][A^-]/[HA]$ and $[H^+] = K_a$ imply $[A^-] = [HA]$. Thus half the dissolved anion remains free.

Step 2: If solubility is s, $[M^{2+}] = s$ and total anion $[A^-] + [HA] = 2s$. Hence free $[A^-] = s$.

Step 3:

$$K_{sp} = [M^{2+}][A^-]^2 = s^3 = 10^{-12}, \quad s = 10^{-4} \text{ M}.$$

Why: Only free ions enter the solubility product; analytical anion concentration includes protonated HA. The two-anion stoichiometry must still be conserved.

Why not the others: (A) It does not satisfy either ion balance or the cubic solubility relation. (B) It takes a square root as for a 1:1 salt. (D) It halves the solved solubility a second time after protonation was already included.

Key point: Couple protonation equilibrium to salt stoichiometry before evaluating Ksp.

Mathematics

Q31 / B Parameter degeneration of a translated conic into crossing lines

Answer: (B) k=0, with intersection (1,1).

Given: The quadratic parts have opposite signs.

Formula/Principle: Complete both squares to locate the centre and detect degeneration.

Step 1:

$$(x-1)^2 - (y-1)^2 + k = 0.$$

Only k=0 makes the right-hand constant vanish in this translated difference of squares.

Step 2: At k=0, factorisation gives

$$(x-y)(x+y-2) = 0.$$

Worked derivation: Solving x=y and x+y=2 yields (1,1).

Why: For nonzero k the translated equation is a nondegenerate hyperbola, not a pair of lines.

Why not the others: (A) The origin is on one line but is not their intersection. (C) k=1 leaves a nonzero hyperbola constant. (D) k=-1 also gives a hyperbola.

Key point: Translate before interpreting the special parameter.

Q32 / A Reconstruct a circle from orthogonality and two incidence constraints

Answer: (A) Centre (1,5/4), radius 5/4.

Given: The squared distance between centres of orthogonal circles equals the sum of their squared radii.

Formula/Principle: With unknown centre (h,k) and radius r, orthogonality gives $h^2+k^2=r^2+1$.

Step 1: Passing through (1,0) gives $r^2=(h-1)^2+k^2$. Subtracting yields h=1.

Step 2: Passing through (0,2) gives $r^2=h^2+(k-2)^2$. Combine with orthogonality: -1=-4k+4, so k=5/4.

Worked derivation:

$$r^2 = h^2 + k^2 - 1 = 25/16, \quad r = 5/4.$$

Why: The tangent lines are perpendicular at an intersection exactly when the radius vectors there are perpendicular, giving the right-triangle centre relation.

Why not the others: (B) It uses centre distance as radius. (C) It fails the (0,2) condition. (D) The proposed circle does not pass through (1,0).

Key point: Orthogonality supplies a radius relation, not a tangency relation.

Q33 / C Unequal inverse endpoints in the area between a monotone graph and its inverse

Answer: (C) (17-5√5)/12.

Given: f is increasing, f(x)>x for 0<x≤1, and g(x)<x for 0<x≤1.

Formula/Principle: The inverse-graph area identity is $\int_0^1 g(y)dy = a - \int_0^a f(x)dx$, where f(a)=1.

Step 1: a²+a=1 gives a=(√5-1)/2. The upper-graph integral is $\int_0^1 (x^2+x)dx = 5/6$.

Step 2: The inverse integral is $a - a^3/3 - a^2/2$. Thus area is $5/6 - a + a^3/3 + a^2/2$.

Step 3: Use a²=1-a and a³=2a-1.

Worked derivation:

$$A = 1 - 5a/6 = \frac{17 - 5\sqrt{5}}{12}.$$

Why: The inverse integral covers y from 0 to 1, so its original-function counterpart stops at a, not at 1. This endpoint mismatch matters.

Why not the others: (A) It retains the wrong sign for the inverse contribution. (B) It approximates the bounded area without evaluating the inverse endpoint. (D) It omits the entire lower-graph area.

Key point: Match inverse endpoints before applying reflection-area identities.

Q34 / D Rational half-angle substitution without a spurious endpoint pole

Answer: (D) 1.

Given: The integrand is continuous, including at π/2.

Formula/Principle: The substitution t=tan(x/2) rationalises sin x and dx.

Step 1: $\sin x = 2t/(1+t^2)$, $dx = 2dt/(1+t^2)$, and endpoints are 0 and 1.

Step 2:

$$I = \int_0^1 \frac{2\,dt}{(1+t)^2}.$$

Worked derivation:

$$I = \left[-\frac{2}{1+t}\right]_0^1 = -1 + 2 = 1.$$

Why: This substitution avoids an apparent endpoint singularity produced by multiplying by 1-sin x; the original integral has no such singularity.

Why not the others: (A) It replaces the denominator by 1. (B) It forgets the transformed denominator. (C) It drops the Jacobian factor 2.

Key point: Transform both the integrand and its endpoints.

Q35 / B Normal-intercept range with a singular excluded parameter

Answer: (B) The intercepts form (1/2,∞), with unattained lower bound 1/2.

Given: t=0 is excluded and would give the vertical normal x=0.

Formula/Principle: For nonzero t the tangent slope is 2t and normal slope is -1/(2t).

Step 1: The normal is

$$y - t^2 = -\frac{x-t}{2t} \;\Rightarrow\; y = -\frac{x}{2t} + t^2 + \tfrac{1}{2}.$$

Step 2: Its y-intercept is b(t)=t²+1/2. For t≠0 this exceeds 1/2, can approach it, and reaches every larger value.

Worked derivation:

$$\{b(t) : t \neq 0\} = (1/2, \infty), \quad \inf b = 1/2.$$

Why: The only stationary parameter of b would be zero, which is absent from the family and has no unique y-intercept as a vertical line.

Why not the others: (A) It includes the excluded endpoint. (C) It omits the normal-line intercept offset. (D) Squares cannot make b negative.

Key point: A limiting parameter need not supply an admissible member of a line family.

Q36 / A Codomain restriction checked for well-definedness before bijection

Answer: (A) The exact range is [-2,∞); the proposed smaller codomain omits actual outputs.

Given: Domain includes x=1.

Formula/Principle: A function into a codomain must have every output inside it; bijectivity then requires injectivity and surjectivity.

Step 1: $f'(x) = 3(x^2 - 1) > 0$ for x>1. The function is strictly increasing on its domain, begins at f(1)=-2 and tends to infinity.

Step 2: Its range is exactly [-2,∞), so choosing that codomain makes it bijective.

Worked derivation:

$$f(1) = -2 \notin [0, \infty),$$

which invalidates the proposed replacement as a map from the unchanged domain.

Why: Codomain restriction must be checked for well-definedness before discussing onto status.

Why not the others: (B) It overlooks negative outputs. (C) It gives a codomain too small for the original domain. (D) The real line contains unused values below -2 and is not the smallest codomain.

Key point: A smaller codomain can invalidate a function rather than merely change surjectivity.

Q37 / A Shifted orthogonal matrix determinant selected by sign and trace

Answer: (A) det X=3.

Given: The transpose equation constrains a shifted matrix.

Formula/Principle: Set Y=I+X. Then $YY^T=4I$; take determinants and use the two-dimensional identity det(I+X)=1+tr X+det X.

Step 1:

$$(I + X)(I + X)^T = I + X + X^T + XX^T = 4I.$$

Step 2: $(\det Y)^2 = \det(4I) = 16$. Positivity selects det Y=4.

Worked derivation:

$$4 = 1 + 0 + \det X \;\Rightarrow\; \det X = 3.$$

Why: The sign condition removes the negative determinant branch; the trace condition then recovers the unshifted determinant.

Why not the others: (D) It confuses signs in the shift identity. (B) It ignores the scale 4 in YY^T. (C) Zero trace does not imply zero determinant.

Key point: Shift a transpose-plus-product equation before taking determinants.

Q38 / C Selection weights that preserve rather than assume conditional independence

Answer: (C) q=1/8 and P(X|S)=4/5.

Given: Selection probabilities reweight four originally independent outcomes.

Formula/Principle: A strictly positive 2×2 probability table represents independence exactly when the products of its opposite corners agree; the common normalisation cancels.

Step 1: The unnormalised selected weights for 11,10,01,00 are

$$1/6, \quad 1/6, \quad 1/24, \quad q/3.$$

Step 2: Independence requires

$$(1/6)(q/3) = (1/6)(1/24) \;\Rightarrow\; q = 1/8.$$

Step 3: The last weight is then 1/24 and P(S)=5/12.

Worked derivation:

$$P(X \mid S) = \frac{1/6 + 1/6}{5/12} = \frac{4}{5}.$$

Why: Original independence alone does not guarantee independence in a selected sample; the selection weights must preserve the table's factorisation.

Why not the others: (A) q=1/4 violates the corner-product equality. (B) It has the correct q but misnormalises the X row. (D) q=0 leaves a missing corner with positive opposing corners and violates independence.

Key point: Check conditional independence using the reweighted joint distribution.

Q39 / D Trigonometric root coalescence at a parameter threshold

Answer: (D) 3,4,2.

Given: The interval includes zero and excludes 2π.

Formula/Principle: Factor the equation and count distinct angular solutions, not algebraic multiplicities.

Step 1: $\sin x(\sin x - a) = 0$ always gives x=0,π.

Step 2: At a=1 the additional root is π/2 only; at a=1/2 there are two additional roots, π/6 and 5π/6; at a=3/2 there are none.

Worked derivation:

$$(N(1), N(1/2), N(3/2)) = (3, 4, 2).$$

Why: At a=1 two solutions of sin x=a coalesce into the double root π/2, but still count as one distinct solution.

Why not the others: (A) It omits the peak root. (B) It omits one of the two half-height roots. (C) It counts the double root twice.

Key point: A tangency changes multiplicity and distinct-root count differently.

Q40 / A Intercept-triangle area combined with distance to its plane

Answer: (A) 18.

Given: Intercepts are (2,0,0),(0,3,0),(0,0,6).

Formula/Principle: Triangle area is half a cross-product magnitude; point-to-plane distance uses the normal's magnitude.

Step 1: The edge cross product is

$$(-2, 3, 0) \times (-2, 0, 6) = (18, 12, 6),$$

so A=3√14.

Step 2: The plane is 3x+2y+z=6, giving d=6/√14.

Worked derivation:

$$Ad = (3\sqrt{14})(6/\sqrt{14}) = 18.$$

Why: The same normal scale occurs in the area and distance and cancels in their product.

Why not the others: (B) It uses only the plane constant. (C) It reports area alone. (D) It omits the triangle's factor 1/2.

Key point: Use the intercept triangle's actual spatial area.

Q41 / C Orientation selects a scalar triple-product sign after norm decomposition

Answer: (C) -2.

Given: a,b,a×b form a right-handed orthonormal basis.

Formula/Principle: Resolve c into this basis and apply the squared-norm identity.

Step 1: Write $c = a + 2b + \gamma(a \times b)$. Its norm gives

$$9 = 1 + 4 + \gamma^2 \;\Rightarrow\; \gamma = \pm 2.$$

Step 2: The specified orientation selects γ=-2.

Worked derivation:

$$a \cdot (b \times c) = (a \times b) \cdot c = \gamma = -2.$$

Why: Length and projections fix the magnitude of the third component but do not fix its sign without orientation information.

Why not the others: (A) It chooses the opposite orientation. (B) It subtracts only one projection square. (D) It mistakes the full vector length for its perpendicular component.

Key point: Signed volume requires orientation as well as lengths.

Q42 / D Reciprocal image of a line with a missing limiting point

Answer: (D) The stated circle with its origin point removed.

Given: No z on Re z=2 is zero, and reciprocal values themselves cannot be zero.

Formula/Principle: Substitute z=1/w and equate real parts.

Step 1:

$$z = \frac{u - iv}{u^2 + v^2}, \qquad \frac{u}{u^2 + v^2} = 2.$$

Step 2: Complete the square:

$$u^2 + v^2 - u/2 = 0 \;\Rightarrow\; (u - 1/4)^2 + v^2 = 1/16.$$

Worked derivation: Every nonzero point on this circle reconstructs a finite z with Re z=2; w=0 does not.

Why: Multiplying by u²+v² introduces the origin into the algebraic equation, so it must be excluded from the actual reciprocal image.

Why not the others: (A) It gives the wrong centre and radius. (B) It retains the spurious zero point. (C) Taking a reciprocal does not preserve the real part as a constant.

Key point: Check excluded points after clearing complex denominators.

Q43 / C Closed-interval positivity versus global quadratic positivity

Answer: (C) a>0.

Given: Positivity is required only on the closed interval, including its endpoints.

Formula/Principle: Locate the quadratic's vertex relative to the interval.

Step 1: At x=0 the value is 2a, so a>0 is necessary. For 0<a<1 the vertex x=a lies inside and its value is $2a - a^2 = a(2 - a) > 0$.

Step 2: For a≥1 the quadratic is decreasing on [0,1], and its minimum is at x=1 with value 1.

Worked derivation:

$$\min_{[0,1]} q = \begin{cases} a(2-a), & 0 < a < 1, \\ 1, & a \geq 1. \end{cases}$$

Both positive-a cases succeed; a≤0 fails at x=0. Thus a>0.

Why: Requiring a negative discriminant would unnecessarily impose positivity on the whole real line.

Why not the others: (A) It excludes valid a≥2. (B) It includes a=0, where the endpoint value is zero. (D) It discards valid 0<a≤2.

Key point: Restrict the minimisation domain before applying a positivity criterion.

Q44 / B Alternating falling-factorial binomial sum evaluated at a nonzero residual base

Answer: (B) -336.

Given: The weight is the third falling factorial, and terms with k<3 vanish.

Formula/Principle: Differentiate the finite binomial polynomial three times.

Step 1:

$$(2-x)^8 = \sum_{k=0}^{8} (-1)^k \binom{8}{k} 2^{8-k} x^k.$$

Step 2: Its third derivative is

$$-8 \cdot 7 \cdot 6 (2-x)^5.$$

Evaluating at x=1 leaves exactly the requested sum because every surviving x power becomes one.

Worked derivation:

$$S = -8 \cdot 7 \cdot 6 = -336.$$

Why: Three derivatives produce k(k-1)(k-2), not k^3; the alternating sign remains in the finite polynomial identity.

Why not the others: (A) It loses the sign of the third derivative. (C) It evaluates at x=2 instead of x=1. (D) It inserts an extra factor 1/2 not present in differentiation.

Key point: Match falling-factorial weights to the number of derivatives.

Q45 / D Finite GP product reconstructed from its central term

Answer: (D) -128.

Given: Seven terms with nonzero real ratio and middle term -2.

Formula/Principle: Terms equally distant from the middle have product equal to the square of the middle term.

Step 1: Pair terms 1 and 7, 2 and 6, 3 and 5. Each pair has product $a_4^2 = 4$, even if the ratio is negative.

Worked derivation:

$$\prod_{j=1}^{7} a_j = 4^3(-2) = -128 = (-2)^7.$$

Why: Symmetric GP exponent sums eliminate the unknown ratio.

Why not the others: (A) It loses the unpaired negative middle term. (B) It omits a factor 2. (C) It multiplies only the three pairs.

Key point: In an odd-length GP the product equals the middle term raised to the number of terms.

Physics

Q46 / A Recover laboratory launch components from two moving-camera observations

Answer: (A) arctan(5/2) and 18 m.

Given: The camera has constant horizontal velocity and no vertical motion.

Formula/Principle: Relative horizontal velocity is ux-2; vertical motion remains y=uy t-gt²/2.

Step 1: From xrel(1)=6, ux=8 m s⁻¹. From y(1)=15, uy-5=15 gives uy=20 m s⁻¹. The second observation checks 40-20=20 m.

Step 2: The launch angle satisfies tan θ=20/8=5/2. For y=15, $20t - 5t^2 = 15$ has roots 1 and 3 s.

Worked derivation:

$$x_{rel}(3) = (8 - 2)3 = 18 \text{ m}.$$

Why: Horizontal camera motion must be restored when finding laboratory angle, but removed again for the requested relative position.

Why not the others: (B) It uses the relative horizontal speed in the laboratory angle. (C) It reports the laboratory coordinate. (D) It uses inconsistent velocity components and the wrong return time.

Key point: Keep laboratory and moving-frame coordinates distinct.

Q47 / A Zero work with a finite impulse from smooth directional reversal

Answer: (A) Zero work and -4 N s impulse.

Given: Speed is constant but velocity reverses.

Formula/Principle: Work equals kinetic-energy change; impulse equals vector momentum change.

Step 1:

$$W = \tfrac{1}{2}(1)(2^2 - 2^2) = 0.$$

Step 2:

$$J_x = m(v_{fx} - v_{ix}) = 1(-2 - 2) = -4 \text{ N s}.$$

Worked derivation: A perpendicular force has $F \cdot v = 0$ throughout while continuously turning the momentum.

Why: Kinetic energy depends on speed squared, whereas momentum retains direction. Zero work therefore does not imply zero impulse.

Why not the others: (B) The speed change is zero, not the velocity change. (C) It ignores reversal of momentum. (D) It assigns energy loss despite unchanged speed.

Key point: Distinguish scalar energy change from vector momentum change.

Q48 / C Off-centre impulse partitions rod energy between translation and rotation

Answer: (C) One quarter.

Given: The impulse produces both translation and rotation.

Formula/Principle: MV=J and Iω=J(L/6); total energy is translational plus centre-of-mass rotational energy.

Step 1:

$$K_{tr} = J^2/(2M).$$

Step 2:

$$K_{rot} = \frac{[J(L/6)]^2}{2(ML^2/12)} = J^2/(6M).$$

Worked derivation:

$$\frac{K_{rot}}{K_{tr} + K_{rot}} = \frac{1/6}{1/2 + 1/6} = 1/4.$$

Why: The external impulse fixes linear momentum, while its lever arm independently fixes angular momentum. Translation cannot be omitted from total energy.

Why not the others: (A) Equal impulse origin does not imply equal energies. (B) Three quarters is the translational fraction. (D) One third is Krot/Ktr rather than Krot/Ktotal.

Key point: Use the same impulse in separate linear and angular momentum balances.

Q49 / B Reduced-gravity capillary rise versus unchanged curvature pressure

Answer: (B) Rise 24 cm, unchanged pressure jump.

Given: Tube geometry and liquid interfacial properties are unchanged.

Formula/Principle: $h = 2\gamma\cos\theta/(\rho g r)$, while the meniscus pressure-jump magnitude is $2\gamma/r$ at zero contact angle.

Step 1: Inverse-gravity scaling gives $h' = 6h = 24$ cm.

Step 2: The curvature set by the same tube gives the same Laplace pressure jump.

Worked derivation:

$$\rho(g/6)(6h) = \rho g h = 2\gamma/r.$$

Why: The taller column compensates weaker gravity; the interfacial curvature does not require a larger pressure difference.

Why not the others: (A) It scales pressure with height but omits the reduced gravity. (C) It ignores hydrostatic balance. (D) It reverses the capillary-height scaling.

Key point: Separate curvature pressure from the height needed to balance it.

Q50 / D Engine-driven refrigeration with a separate parasitic heat leak

Answer: (D) 2600 J net removed from the chamber and 3800 J net gained by the room.

Given: The engine and refrigerator are reversible; the heat leak is separate.

Formula/Principle: Engine efficiency is 1-Tc/Th; refrigerator COP is Tcold/(Thot-Tcold).

Step 1:

$$W = 1200(1 - 300/600) = 600\ \text{J}, \quad Q_{engine,room} = 600\ \text{J}.$$

Step 2: Refrigerator COP=250/50=5, so it removes 3000 J and rejects 3600 J to the room.

Step 3: The leak reduces net cold removal to 3000-400=2600 J and reduces room gain by the same 400 J.

Worked derivation:

$$Q_{room,net} = 600 + 3600 - 400 = 3800\ \text{J} = 1200 + 2600.$$

Why: The room is both a heat sink for two devices and the source of the parasitic leak; gross refrigerator cooling is not net cooling.

Why not the others: (A) It ignores the leak in both budgets. (B) It includes the leak only in the chamber budget. (C) It misuses the engine work or refrigerator COP.

Key point: Track each thermal reservoir separately and verify total energy balance.

Q51 / C Common standing-wave frequency across a nodal density junction

Answer: (C) Frequency √(T/μ)/(2L), with four distinct nodes.

Given: The junction is specified to be a node.

Formula/Principle: Each segment is node-bounded, so f=nivi/(2L); wave speed is √(T/μi).

Step 1: If the light-segment speed is v, the heavy-segment speed is v/2. Equality of frequency requires nheavy=2nlight. Lowest integers are 1 and 2.

Step 2: The light segment has only its endpoints as nodes; the heavy segment adds one internal node.

Worked derivation:

$$f = v/(2L), \quad N_{nodes} = 2 + 3 - 1 = 4.$$

Why: Frequency is common across the joined mode, while wavelengths and harmonic indices differ. The shared junction is counted once.

Why not the others: (A) Its frequency cannot give an integer light-segment harmonic. (B) It is a higher common mode. (D) It misses the heavy segment's internal node.

Key point: Match frequency rather than wavelength across different string densities.

Q52 / B Global field maximum for a uniform core plus a separated shell

Answer: (B) At radius R, with magnitude GM/R².

Given: Both distributions are spherical.

Formula/Principle: Enclosed mass determines field; a spherical shell has zero field inside.

Step 1: Inside the core, $g(r) = GMr/R^3$, increasing to GM/R². Between R and 2R, the shell contributes zero and the core field decreases as 1/r².

Step 2: Just outside the shell, field is $2GM/(2R)^2 = GM/(2R^2)$, below the core-surface maximum, and decreases farther out.

Worked derivation:

$$g(R) = GM/R^2 > g(2R^+) = GM/(2R^2).$$

Why: The shell adds mass only to the exterior field, not to the field at the smaller core surface.

Why not the others: (A) It identifies a local upward jump but not the global maximum. (C) Symmetry makes the centre field zero. (D) It includes the shell as if its mass were enclosed at R.

Key point: Compare all radial regions when a shell causes a field jump.

Q53 / A Axis-constrained stability in a quadrupole with a neutral perpendicular axis

Answer: (A) Restoring x, unstable y, neutral z.

Given: Opposite-axis charge pairs have equal magnitudes; all charges are fixed.

Formula/Principle: Test-charge force is -Q times the potential gradient; potential curvature determines local restoring behaviour.

Step 1: Along x,

$$V(x) = kq\left(\frac{1}{a-x} + \frac{1}{a+x} - \frac{2}{\sqrt{a^2+x^2}}\right).$$

Its quadratic term is $3kqx^2/a^3$, hence $F_x \simeq -6kqQx/a^3$.

Step 2: Interchanging x,y reverses charge signs, so the y force is anti-restoring. On the z-axis all four distances are equal and the pair fields cancel exactly.

Worked derivation:

$$V_{xx}(0) = 6kq/a^3 > 0, \quad V_{yy}(0) = -6kq/a^3 < 0, \quad E(0,0,z) = 0.$$

Why: Zero field establishes equilibrium, not stability in every allowed direction. The curvatures can also be obtained by twice differentiating the displayed potential, without a series-convergence assumption.

Why not the others: (B) It reverses the two charged-axis curvatures. (C) A potential value does not determine its curvature. (D) It extends one-point field cancellation to nearby x and y.

Key point: State the allowed displacement direction when assessing electrostatic equilibrium.

Q54 / D Sequential pair sharing with an isolated intermediate capacitor

Answer: (D) V/2 and 5C0V²/16.

Given: Only the connected pair can exchange charge at each stage.

Formula/Principle: Pair charge is conserved; resistive heat is the decrease in total electrostatic energy, with no battery work.

Step 1: After the first connection A and B are both at V/2. B remains at that voltage thereafter.

Step 2: In the second connection A's charge C0V/2 divides equally between A and C, leaving each at V/4.

Step 3: Initial energy is C0V²/2; final energy is

$$U_f = \tfrac{1}{2}C_0[(V/2)^2 + 2(V/4)^2] = 3C_0V^2/16.$$

Worked derivation:

$$Q_{heat} = U_i - U_f = 5C_0V^2/16.$$

Why: Sequential isolation prevents the all-three common-voltage result; the charge-conservation subsystem changes between stages.

Why not the others: (A) It assumes all three remain connected together. (B) It gives B the voltage of the second-stage pair. (C) It counts first-stage heat only.

Key point: Redraw which capacitors can exchange charge after every switch.

Q55 / A Load threshold reverses current in a weaker parallel cell

Answer: (A) Threshold 1 ohm; charging current 6/5 A at R=2 ohm.

Given: The common terminal voltage is U.

Formula/Principle: Branch currents delivered by cells are (E-U)/r; a negative value denotes charging.

Step 1: Kirchhoff's current law gives

$$(12 - U) + (6 - U) = U/R \;\Rightarrow\; U = 18R/(2R + 1).$$

Step 2: Zero weak-cell current requires U=6 V, giving 18R=12R+6 and R=1 ohm.

Step 3: At R=2, U=36/5 V.

Worked derivation:

$$I_{charge} = (U - 6)/1 = 6/5 \text{ A}.$$

Why: For R>1 the stronger cell raises terminal voltage above the weak cell's emf, driving current into its positive terminal.

Why not the others: (B) R=2 is the evaluation load, not the threshold. (C) 18/5 A is the load current at R=2. (D) Neither value satisfies the terminal-voltage equation.

Key point: Determine cell charging from terminal voltage relative to its emf.

Q56 / B Magnetic-stripe width decides return rather than transmission

Answer: (B) It returns through the left boundary after half a cyclotron period, moving -x.

Given: The stripe is wider than the orbit's maximum x penetration.

Formula/Principle: The magnetic force turns velocity clockwise in the xy-plane with angular speed Ω=qB/m.

Step 1: Taking the entry point as origin,

$$x = R\sin\Omega t, \quad y = R(\cos\Omega t - 1).$$

Maximum x is R, less than d=1.5R, so it never reaches the right boundary.

Step 2: The first subsequent x=0 occurs at Ωt=π. Its velocity is $(v\cos\pi, -v\sin\pi) = (-v, 0)$.

Worked derivation:

$$t = \pi m/(qB), \quad \mathbf{v}_{exit} = -v\hat{x}.$$

Why: Finite-field trajectories must be checked against the boundary before assuming transmission.

Why not the others: (A) The right boundary lies beyond maximum penetration. (C) It also assumes impossible transmission. (D) The particle exits after half a turn, not a full turn.

Key point: Compare magnetic radius with stripe width to decide exit topology.

Q57 / C Capacitor-voltage peak differs from series-current resonance

Answer: (C) 5√2 rad s⁻¹, with gain 2/√3.

Given: The measured response is capacitor voltage, not circuit current.

Formula/Principle:

$$\left(\frac{V_C}{E}\right)^2 = \frac{1}{(1-\omega^2 LC)^2 + \omega^2 R^2 C^2}.$$

Step 1: Set u=ω² and minimise the denominator D(u). Its derivative is $-2LC(1-uLC) + R^2C^2$.

Step 2: The positive minimum is at

$$u = 1/(LC) - R^2/(2L^2) = 100 - 50 = 50.$$

Step 3: At this point $1 - uLC = 1/2$ and $uR^2C^2 = 1/2$.

Worked derivation:

$$\omega = 5\sqrt{2}, \quad D = 1/4 + 1/2 = 3/4, \quad V_C/E = 2/\sqrt{3}.$$

Why: Capacitive reactance itself varies with frequency; maximising I alone therefore does not maximise IXC. The minimum is interior and below the zero-frequency denominator 1.

Why not the others: (A) It substitutes current-resonance frequency and misses the larger capacitor response. (B) The gain at 10 rad s⁻¹ is 1, not √2. (D) It does not satisfy the derivative condition.

Key point: Specify which circuit response is being maximised.

Q58 / B Fixed-lens microscope refocusing from infinity to a finite virtual image

Answer: (B) New distance 41/35 cm; movement 1/35 cm toward the objective.

Given: Paraxial thin lenses, with a real intermediate image to the left of the eyepiece.

Formula/Principle: Use 1/f=1/v-1/u for each lens and impose the fixed separation.

Step 1: Initially the intermediate image is 5 cm before the eyepiece, hence 6 cm after the objective. Initial object distance is $6/(6-1) = 6/5$ cm.

Step 2: For the eyepiece v=-25 cm:

$$1/5 = -1/25 - 1/u_e \;\Rightarrow\; u_e = -25/6 \text{ cm}.$$

The objective image distance becomes $11 - 25/6 = 41/6$ cm.

Step 3: The new objective object distance is

$$(41/6)/(41/6 - 1) = 41/35 \text{ cm}.$$

Worked derivation:

$$6/5 - 41/35 = 1/35 \text{ cm},$$

a decrease, so movement is toward the objective.

Why: Final-image adjustment first changes the required intermediate image; objective conjugacy then fixes the object displacement.

Why not the others: (A) It retains the old object distance. (C) It reverses the displacement direction. (D) It inverts the objective conjugate relation.

Key point: Work backward through the eyepiece before moving the microscope object.

Q59 / D Counterpropagating electromagnetic fields and failure of a single-wave field ratio

Answer: (D) Electric fields add and magnetic fields cancel; their sum generally does not satisfy the single-travelling-wave relation.

Given: Counterpropagating waves have opposite magnetic signs for the same electric direction.

Formula/Principle: Superpose electric and magnetic fields separately. The ratio E=cB describes each individual plane travelling wave.

Step 1: At x=0 both cosine factors equal cos ωt, giving $E_{tot} = 2E_0 \cos\omega t\,\hat{y}$.

Step 2: The two magnetic contributions are equal and opposite.

Worked derivation:

$$\mathbf{E}_{tot} = 2E_0 \cos\omega t\,\hat{y}, \qquad \mathbf{B}_{tot} = 0.$$

Whenever cos ωt is nonzero, the electric magnitude is nonzero while c times the magnetic magnitude is zero.

Why: A superposition of opposite travelling waves need not itself be one travelling wave. Apply the single-wave field relation to the constituent waves before adding their signed vectors.

Why not the others: (A) It swaps the adding and cancelling fields. (B) It incorrectly imposes the single-wave relation on the sum. (C) The electric fields reinforce rather than cancel.

Key point: Superpose vector fields before classifying the combined wave.

Q60 / C Atomic-to-nuclear mass conversion before the positron threshold

Answer: (C) The bare-nucleus mass difference is 0.5 me and positron emission is energetically impossible.

Given: Atomic mass difference is 1.5 me and daughter atomic number is one smaller.

Formula/Principle: Nuclear mass equals atomic mass minus Z electron masses; positron creation costs another me.

Step 1:

$$\Delta M_N = (M_P - Zm_e) - (M_D - (Z-1)m_e) = \Delta M_{atom} - m_e = 0.5m_e.$$

Step 2: Subtract the emitted positron rest mass in the nuclear reaction.

Worked derivation:

$$Q_{\beta+} = (\Delta M_N - m_e)c^2 = (0.5-1)m_e c^2 = -0.5m_e c^2 < 0.$$

Why: The parent atom contains one more electron than the daughter. Converting the two atomic masses to nuclei removes one electron mass from their difference before the positron mass is counted.

Why not the others: (A) It omits both corrections. (B) It adds the atomic electron correction with the wrong sign. (D) A nuclear gap below one positron rest mass cannot supply its creation energy.

Key point: Match electron bookkeeping to the mass convention before evaluating nuclear Q.
